

Loan Utilization Tracking System

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ABSTRACT :

- The Loan Utilization Tracking System is a web and mobile-based application designed to monitor how loan funds are used by beneficiaries.
- It replaces manual verification with a digital system where users upload geo-tagged and time-stamped asset proofs.
- Authorities can remotely verify submissions, reducing the need for physical inspections.
- Built using the MERN stack and Cloudinary, the system improves transparency, reduces fraud, and enhances efficiency in loan monitoring.

INTRODUCTION

- Traditional loan verification methods are slow, costly, and depend on manual field visits, making the process inefficient.
- Manual monitoring of loan utilization lacks accuracy, transparency, and proper tracking, leading to misuse of funds.
- There is a growing need for smart digital solutions to monitor and verify loan usage in real time.
- The system is developed as a Web & Mobile-Based AI-Powered Financial Monitoring System using the MERN stack.
- It enables geo-tagged proof uploads, remote verification, and automated monitoring to reduce fraud and improve efficiency.

PROBLEM STATEMENT:

Traditional loan verification methods are time-consuming, costly, and rely on manual field visits. There is a lack of real-time monitoring and transparency in tracking loan utilization, leading to fund misuse and fraud. Therefore, a digital system is needed to enable efficient verification, remote monitoring, and accurate tracking of loan usage.

OBJECTIVES

- To design and develop a digital system for tracking loan utilization by beneficiaries.
- To enable users to upload geo-tagged and time-stamped proof of asset purchases.
- To provide real-time monitoring and verification of loan usage.
- To reduce fraud and misuse of loan funds through digital validation.
- To allow administrators to review, approve, or reject submitted proofs efficiently.
- To improve transparency, accountability, and efficiency in loan management processes.

EXISTING SYSTEM

- Manual loan monitoring using paperwork, spreadsheets, and field visits
- Traditional verification through physical inspection by officers
- Limited use of digital systems for tracking loan utilization
- Separate and unconnected systems for loan data, verification, and reporting
- No proper real-time monitoring or automated tracking mechanisms

PROPOSED SYSTEM

Proposed System:

- A web and mobile-based system to digitally track loan utilization
- Beneficiaries upload geo-tagged and time-stamped asset proofs
- Authorities verify submissions remotely through an admin dashboard
- Reduces manual work, physical inspections, and fraud
- Provides real-time monitoring and improves transparency

Key Features:

- Mobile login
- Proof upload (image/video)
- Cloud storage (Clouinary)
- Admin verification
- Real-time updates

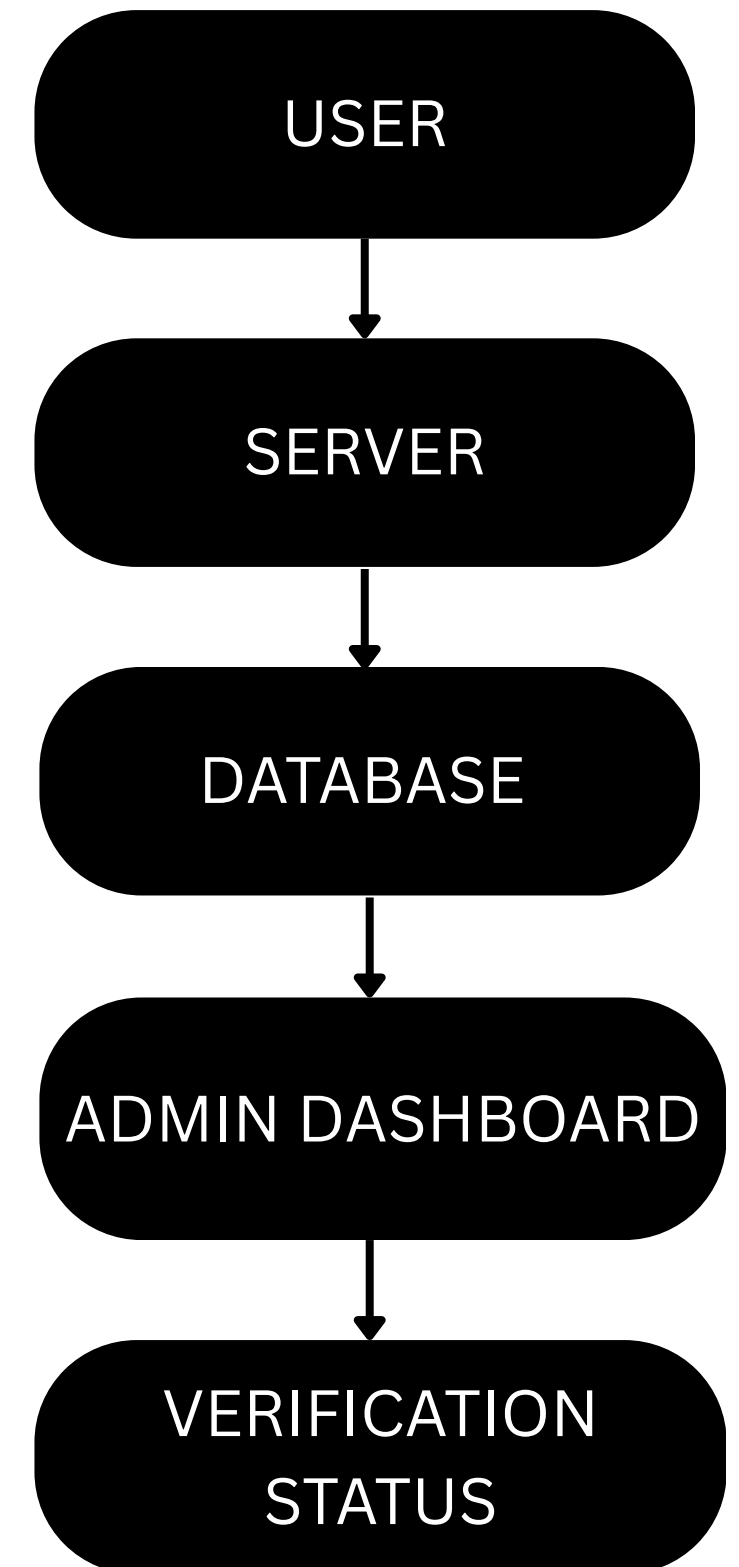
WORKING PROCESS

1. User registers/logs into the system using mobile number
2. Admin enters loan details and assigns beneficiary
3. User uploads geo-tagged and time-stamped asset proof
4. Data is stored in database and media stored in Cloudinary
5. Admin reviews uploaded proof on dashboard
6. Verification done and status updated (approved/rejected)

SYSTEM ARCHITECTURE

- User Interface (React)
- Backend Server (Node.js + Express)
- Database (MongoDB)
- Cloud Storage (Cloudinary for images/videos)
- Admin Dashboard (for verification & monitoring)
- AI Verification Module (for asset validation)
- Real-Time Notification System

DATA FLOW



TOOLS & TECHNOLOGIES

FRONTEND	React.js Tailwind Css
BACKEND	Node.js Express.js
DATABASE	MongoDB
TOOLS	VS Code GitHub
STORAGE	Cloudinary

FEASIBILITY STUDY

TECHNICAL FEASIBILITY

- Uses modern MERN stack (MongoDB, Express.js, React, Node.js)
- Easy integration with cloud storage (Clouldinary)
- Supports mobile and web platforms
- Feasible with existing technologies and tools
- Low hardware and infrastructure requirements
- Development can be completed within 4–6 months
- Cost-effective and scalable solution

EXPECTED OUTCOMES

- Efficient digital tracking of loan utilization by beneficiaries
- Reduction in fraud and misuse of loan funds
- Real-time monitoring and faster verification process
- Improved transparency and accountability in loan management
- Reduced need for physical inspections and manual work
- Better data management and accurate record keeping
- Enhanced decision-making for authorities

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THANK
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